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Bereken $\int \frac{4}{9^x} dx$

$$\int \frac{4}{9^x} dx = 4 \int 9^{-x} dx$$

Substitueer $t = -x$, dan is $dt = -dx$

$$\begin{aligned} 4 \int 9^{-x} dx &= -4 \int 9^t dt \\ &= -4 \cdot \frac{9^t}{\ln 9} + c \\ &= \frac{-4 \cdot 9^{-x}}{\ln 9} + c \\ &= \frac{-4}{9^x \cdot \ln 9} + c \end{aligned}$$

References